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(54) **NOVEL COMPOUNDS FOR THE TREATMENT OF PARASITIC INFECTIONS**

NEUARTIGE VERBINDUNGEN ZUR BEHANDLUNG PARASITÄRER INFEKTIONEN

NOUVEAUX COMPOSÉS POUR LE TRAITEMENT D'INFECTIONS PARASITAIRES

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WO-A1-2013/121387 WO-A1-2017/067881

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Description**Field of the invention**

[0001] The present invention relates to novel compounds or pharmaceutically acceptable salts thereof, corresponding compositions, and methods and/or uses in therapy, for example in the treatment of parasitic infections such as malaria, in particular infection by *Plasmodium falciparum*.

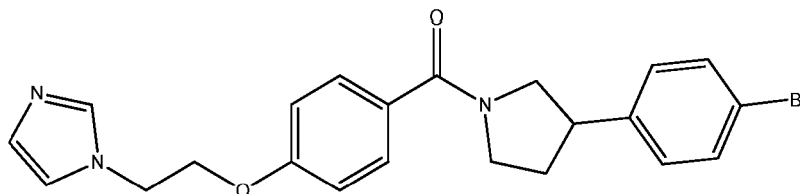
Background of the invention

[0002] Parasitic protozoal infections are responsible for a wide variety of diseases of medical and veterinary importance, including malaria in man and various coccidiosis in birds, fish and mammals. Many of the diseases are life-threatening to the host and cause considerable economic loss in animal husbandry, such as diseases caused by infection by species of *Eimeria*, *Theileria*, *Babesia*, *Cryptosporidium*, *Toxoplasma* (such as *Toxoplasma brucei*, African sleeping sickness and *Toxoplasma cruzi*, Chagas disease) and *Plasmodium* (such as *Plasmodium falciparum*), and the Mastigophora such as species of *Leishmania* (such as *Leishmania donovani*). Another parasitic organism of increasing concern is *Pneumocystis carinii*, which can cause an often fatal pneumonia in immunodeficient or immunocompromised hosts, including those infected with HIV.

[0003] Malaria is a mosquito-borne disease that, in humans, can be caused by five species of *Plasmodium* parasite, of which *Plasmodium falciparum* is the most virulent. In 2016, there were an estimated 216 million of people infected with malaria worldwide and malarial disease was responsible for an estimated 445,000 deaths (91% of them in sub-Saharan Africa), young children and pregnant women being the most affected groups (WORLD HEALTH ORGANIZATION. (2017. *World malaria report*. Geneva, Switzerland, World Health Organization).

[0004] Resistance to classical treatments and emerging resistance to the current treatment of choice (artemisinin-based combination therapies) reveals the urgent need for new therapeutic agents with novel mechanisms of action (WORLD HEALTH ORGANIZATION. Joint assessment of the response to artemisinin resistance in the greater Mekong sub-region. November 2011-February 2012. Summary report).

[0005] In 2010, GSK released details of more than 13,500 chemical compounds that have already shown to inhibit *Plasmodium falciparum* parasite growth in the phenotypic screening approach (Gamo, F.J. et al., Thousands of chemical starting points for antimalarial lead identification. *Nature*, 2010, 465, 305-310). Molecular structures and descriptions of these compounds were made publicly available in accessible databases under the name of TCAMS (Tres Cantos Antimalarial set) (<http://www.ebi.ac.uk/chemblntd>). One such compound disclosed in TCAMS is:



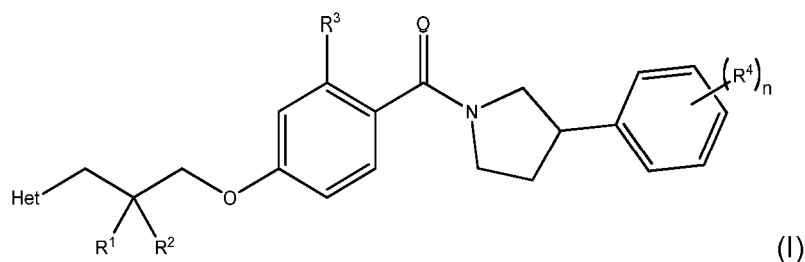
[0006] WO2017/067881 discloses substituted pyrazines for use in the treatment of parasitic protozoal infections.

[0007] Owing to the ever growing emergence of drug resistant strains of malaria and continued high incidence of malaria, there exists an urgent need to provide further drug compounds for the treatment of malaria.

Summary of the invention

[0008] Any references to methods of treatment refer to the compounds, pharmaceutical compositions and medicaments of the present invention for use in a method for treatment of the human (or animal) body by therapy (or for diagnosis).

[0009] In a first aspect of the invention, there is provided a compound of Formula (I) or pharmaceutically acceptable salt thereof:



wherein

R¹ is hydrogen, fluoro or methyl, wherein

when R¹ is hydrogen, R² is fluoro, methyl, hydroxyl or amino,

when R¹ is fluoro, R² is fluoro, and

when R¹ is methyl, R² is methyl or hydroxyl;

R³ is hydrogen or fluoro;

R⁴ is fluoro, chloro, cyano or methoxy;

n is 1 or 2; and

Het is imidazole, pyrazole, thiazole, isothiazole, oxazole, isoxazole, triazole or tetrazole, wherein each Het is optionally substituted by C₁-C₄ alkyl, cyano or -C(O)NH₂.

[0010] In a second aspect of the invention, there is provided a pharmaceutical composition comprising (a) a compound of Formula (I) or pharmaceutically acceptable salt thereof; and (b) a pharmaceutically acceptable excipient.

[0011] In a third aspect of the invention, there is provided a compound of Formula (I) or pharmaceutically acceptable salt thereof, for use in therapy.

[0012] In a fourth aspect of the invention, there is provided a compound of Formula (I) or a pharmaceutically acceptable salt thereof, for use in the treatment of malaria.

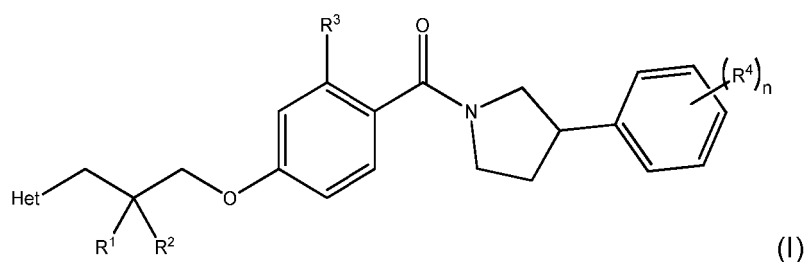
[0013] In a fifth aspect of the invention, there is provided a method for the treatment of a parasitic protozoal infection, comprising administering a pharmaceutically effective amount of a compound of Formula (I) or pharmaceutically acceptable salt thereof to a human in need thereof.

[0014] In a sixth aspect of the invention, there is provided the use of a compound of Formula (I) or pharmaceutically acceptable salt thereof in the manufacture of a medicament for use in the treatment of a parasitic protozoal infection.

[0015] In a seventh aspect of the invention, there is provided a combination of (a) a compound of Formula (I) or pharmaceutically acceptable salt thereof; and (b) at least one other anti-malarial agent.

Detailed description of the invention

[0016] As described above, in one aspect of the invention, there is provided a compound of Formula (I) or a pharmaceutically acceptable salt thereof:



wherein

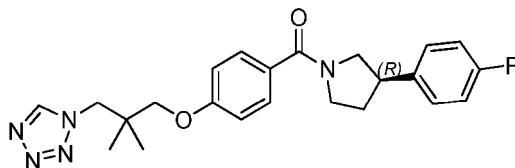
R¹ is hydrogen, fluoro or methyl, wherein

when R¹ is hydrogen, R² is fluoro, methyl, hydroxyl or amino,

when R¹ is fluoro, R² is fluoro, and

Example 2: (R)-(4-(2,2-Dimethyl-3-(1H-tetrazol-1-yl)propoxy)phenyl)(3-(4-fluorophenyl) pyrrolidin-1-yl)methanone

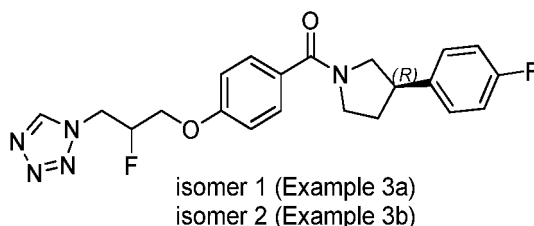
[0292]



[0293] To a solution of 4-(2,2-dimethyl-3-(1H-tetrazol-1-yl)propoxy)benzoic acid (intermediate **56**, 88 mg, 0.319 mmol), (R)-3-(4-fluorophenyl)pyrrolidine hydrochloride (Intermediate **3b**, 70.7 mg, 0.350 mmol) and DIPEA (SIGMA-ALDRICH, 0.167 mL, 0.956 mmol) in DMF (5 mL) at 0°C under nitrogen atmosphere was added COMU (ALFA AESAR, 177 mg, 0.414 mmol) and the reaction mixture was stirred at 0°C for 1 h and then at rt overnight. The mixture was quenched by the dropwise addition of water, and then partitioned between EtOAc and sat. aq. NH₄Cl. The phases were separated and the aqueous one was re-extracted with EtOAc. The organic layers were combined and washed with sat. aq. NH₄Cl, sat. NaHCO₃ and brine. The organic layer was dried over anh. Na₂SO₄, filtered and evaporated to dryness under vacuum. The crude was purified by chromatography on silica gel and eluted with CyHex:EtOAc/EtOH (3/1) (from 100:0 to 30:70) to give the title compound (129 mg, 96%). ¹H NMR (300 MHz, CDCl₃) δ ppm: 8.47 (s, 1H), 7.61-7.52 (m, 2H), 7.26-7.14 (m, 2H), 7.08-6.97 (m, 2H), 6.95-6.86 (m, 2H), 4.54-4.56 (m, 2H), 4.15-3.29 (m, 7H), 2.44-2.25 (m, 1H), 2.18-1.94 (m, 1H), 1.14 (s, 6H). [ES+ MS] m/z 424 (MH⁺).

Examples 3a-b: (4-(2-Fluoro-3-(1H-tetrazol-1-yl)propoxy)phenyl)((R)-3-(4-fluorophenyl) pyrrolidin-1-yl)methanone ((isomer 1 and isomer 2)

[0294]



[0295] To a solution of 4-(2-fluoro-3-(1H-tetrazol-1-yl)propoxy)benzoic acid (intermediate **70**, 145 mg, 0.545 mmol), (R)-3-(4-fluorophenyl)pyrrolidine hydrochloride (intermediate **3b**, 132 mg, 0.654 mmol) and DIPEA (SIGMA-ALDRICH, 0.285 mL, 1.634 mmol) in anh. DMF (10 mL) at 0°C under nitrogen atmosphere was added COMU (ALFA AESAR, 303 mg, 0.708 mmol) and the reaction mixture was stirred at 0°C for 1 h and then at rt overnight. The mixture was quenched by the dropwise addition of water. The resulting mixture was then partitioned between EtOAc and sat. aq. NH₄Cl. The phases were separated and the aqueous one was re-extracted with EtOAc. The organic layers were combined and washed with sat. aq. NH₄Cl, sat. NaHCO₃ and brine. The organic layer was dried over anh. Na₂SO₄, filtered and evaporated to dryness under vacuum. The crude was purified by chromatography on silica gel and eluted with Cy-Hex:EtOAc/EtOH(3/1) (from 100:0 to 40:60) to give a mixture of diastereoisomers (235 mg, quant. yield) that was separated by chiral SFC separation (ChiralPak IC 250X20 mm column, CO₂/MeOH (1%DEA) 60:40 (v/v), F= 70 mL/min). The desired fractions were concentrated under reduced pressure and the residues were dissolved in CH₃CN/ H₂O and lyophilised to give isomer 1 (Example 3a, 87 mg, ee 100%) and isomer 2 (Example 3b, 90 mg, ee 100%).

isomer 1 (Example 3a) - First eluted in SFC. ¹H NMR (300 MHz, CDCl₃) δ ppm: 8.77 (br.s, 1H), 7.64-7.56 (m, 2H), 7.30-7.14 (m, 2H), 7.12-6.90 (m, 3H), 5.35-5.14 (m, 1H), 5.04-4.90 (m, 2H), 4.39-4.07 (m, 3H), 3.98-3.30 (m, 5H), 2.50-2.27 (m, 1H), 2.20-1.94 (m, 1H). [ES+ MS] m/z 414 (MH⁺).

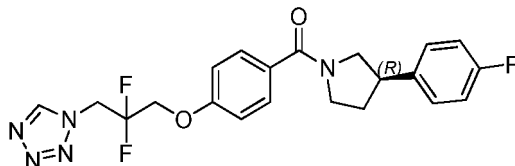
isomer 2 (Example 3b) - Second eluted in SFC. ¹H NMR (300 MHz, CDCl₃) δ ppm: 8.77 (br.s, 1H), 7.64-7.56 (m, 2H), 7.30-7.14 (m, 2H), 7.12-6.90 (m, 3H), 5.35-5.14 (m, 1H), 5.04-4.90 (m, 2H), 4.39-4.07 (m, 3H), 3.98-3.30 (m, 5H), 2.50-2.27 (m, 1H), 2.20-1.94 (m, 1H). [ES+ MS] m/z 414 (MH⁺).

[0296] Chiral SFC determination using ChiralPak IC 150x3 mm column (CO₂/MeOH (1%DEA) 80:20 (v/v), F=2.0

mL/min, R_t isomer 1= 5.58 min, R_t isomer 2= 6.32 min).

Example 4: (R)-(4-(2,2-Difluoro-3-(1H-tetrazol-1-yl)propoxy)phenyl)(3-(4-fluorophenyl)-pyrrolidin-1-yl)methanone

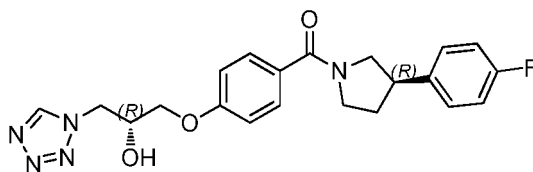
[0297]



[0298] AcOH (SIGMA-ALDRICH, 3.44 mL) was added to a mixture of (R)-(4-(3-amino-2,2-difluoropropoxy)phenyl)(3-(4-fluorophenyl)pyrrolidin-1-yl)methanone (Intermediate **9**, 1.3 g, 3.44 mmol), NaN_3 (SIGMA-ALDRICH, 0.335 g, 5.15 mmol) and triethyl orthoformate (SIGMA-ALDRICH, 2.076 mL, 13.74 mmol). The resulting mixture was refluxed for 20 h. The reaction mixture was cooled to rt, and solvent was removed under reduced pressure. The crude material was dissolved in EtOAc, washed with sat. NH_4Cl solution, dried over Na_2SO_4 , filtered, and concentrated under reduced pressure. The crude was purified by chromatography on silica gel and eluted with CyHex:EtOAc/EtOH (3/1) (from 100:0 to 50:50) to obtain the title compound (287 mg, 19.4%). ^1H NMR (300 MHz, CDCl_3) δ ppm: 8.77 (s, 1H), 7.62-7.54 (m, 2H), 7.26-7.12 (m, 2H), 7.08-6.88 (m, 4H), 5.17-5.05 (m, 2H), 4.25 (m, 2H), 4.15-3.32 (m, 5H), 2.45-2.25 (m, 1H), 2.18-1.91 (m, 1H). [ES+ MS] m/z 432 (MH^+).

Example 5: ((R)-3-(4-Fluorophenyl)pyrrolidin-1-yl)(4-((R)-2-hydroxy-3-(1H-tetrazol-1-yl) propoxy)phenyl)methanone

[0299]

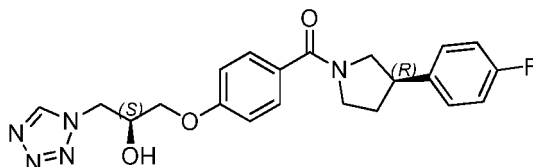


[0300] To a solution of (R)-4-(2-hydroxy-3-(1H-tetrazol-1-yl)propoxy)benzoic acid (intermediate **60**, 1 g, 3.78 mmol), (R)-3-(4-fluorophenyl)pyrrolidine hydrochloride (Intermediate **3b**, 0.916 g, 4.54 mmol) and DIPEA (SIGMA-ALDRICH, 1.983 mL, 11.35 mmol) in anh. DMF (37.8 mL) at 0°C under nitrogen atmosphere was added COMU (ALFA AESAR, 2.107 g, 4.92 mmol) and the reaction mixture was stirred at 0°C for 1 h and then at rt overnight. The mixture was quenched by the dropwise addition of water, and then partitioned between EtOAc and sat. aq. NaHCO_3 .

[0301] The phases were separated and the aqueous one was re-extracted with EtOAc. The organic layers were combined and washed with sat. NaHCO_3 , 1 N aq. NH_4Cl and brine. The organic layer was dried over anh. Na_2SO_4 , filtered and evaporated to dryness under vacuum. The crude was purified by chromatography on silica gel and eluted with CyHex:EtOAc/EtOH (3/1) (from 100:0 to 50:50). The desired fractions were collected and the solvent was removed in vacuo to obtain the desired product as a dark red oil. It was dissolved in EtOAc and washed with 2 N HCl twice. The organic layer was dried over anh. Na_2SO_4 , filtered and evaporated to dryness under vacuum to obtain the desired product as a colourless oil. It was lyophilized to obtain the title compound (1.38 g, 89 % yield, 95% purity by chiral SFC analysis, ee 90%) as a white solid. The title compound was purified by chiral SFC (ChiralPak IA 250X20 mm column, CO_2/MeOH (1%DEA) 70:30 (v/v), $F=80$ mL/min). The desired fractions were concentrated under reduced pressure and the residue was dissolved in EtOAc and then washed with sat. aq. NaHCO_3 . Organic layer was dried over anh. MgSO_4 , evaporated to dryness under vacuum and the residue was lyophilized to obtain the title compound (656 mg, ee 100%). ^1H NMR (300 MHz, CDCl_3) δ ppm: 8.85 (d, $J=5.3$ Hz, 1H), 7.53-7.43 (m, 2H), 7.29-7.13 (m, 2H), 7.11-6.99 (m, 2H), 6.81 (dd, $J=11.4, 8.6$ Hz, 2H), 4.98 (br.s, 1H), 4.83-4.74 (m, 1H), 4.63-4.52 (m, 1H), 4.22 (br.s, 1H), 3.92-3.32 (m, 7H), 2.46-2.27 (m, 1H), 2.20-1.96 (m, 1H). [ES+ MS] m/z 412 (MH^+). Chiral SFC determination using ChiralPak IA 150x3 mm column (CO_2/EtOH (1%DEA) 60:40 (v/v), $F=3.0$ mL/min, $R_t=2.70$ min).

Example 6: ((R)-3-(4-Fluorophenyl)pyrrolidin-1-yl)(4-((S)-2-hydroxy-3-(1H-tetrazol-1-yl) propoxy)phenyl)methanone

[0302]



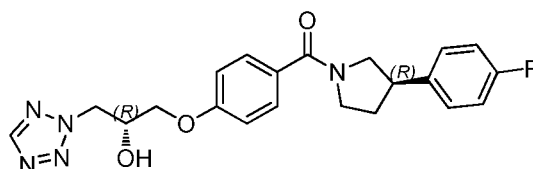
[0303] The title compound was prepared using analogous conditions described in Example 5 using (S)-4-(2-hydroxy-3-(1H-tetrazol-1-yl)propoxy)benzoic acid (intermediate **62**, 350 mg, 1.32 mmol). 256 mg (ee 100%) of the title compound as a white solid was obtained after chiral purification. ¹H NMR (300 MHz, CDCl₃) δ ppm: 8.85 (d, J= 5.3 Hz, 1H), 7.56-7.46 (m, 2H), 7.29-7.14 (m, 2H), 7.12-6.99 (m, 2H), 6.84 (dd, J= 11.4, 8.6 Hz, 2H), 4.86-4.74 (m, 1H), 4.68-4.55 (m, 1H), 4.34 (br.s, 1H), 4.19 (br.s, 1H), 3.92-3.32 (m, 7H), 2.48-2.27 (m, 1H), 2.20-1.95 (m, 1H). [ES+ MS] m/z 412 (MH⁺). Chiral SFC determination using ChiralPak IA 150x3 mm column (CO₂/MeOH (1%DEA) 70:30 (v/v), F= 3.0 mL/min, R_t = 6.72 min).

[0304] Example 5 and Example 6 were also prepared using analogous conditions described in Example 1a-b, using (4-(2-hydroxy-3-(1H-tetrazol-1-yl)propoxy)benzoic acid (intermediate **59**, 85 mg, 0.322 mmol). The diastereomer mixture was separate by chiral SFC to afford isomer 1 (17 mg, ee 100%, that correspond with Exampe 6 - S isomer) and isomer 2 (11 mg, ee 100%, that correspond with Example 5 - R isomer).

[0305] Chiral SFC determination using ChiralPak IA 150x3 mm column (CO₂/MeOH (1%DEA) 70:30 (v/v), F=3.0 mL/min, R_t isomer 1 (example6) = 5.8 min, R_t isomer 2 (example5) = 8.0 min).

Example 7: ((R)-3-(4-Fluorophenyl)pyrrolidin-1-yl)(4-((R)-2-hydroxy-3-(2H-tetrazol-2-yl) propoxy)phenyl)methanone

[0306]

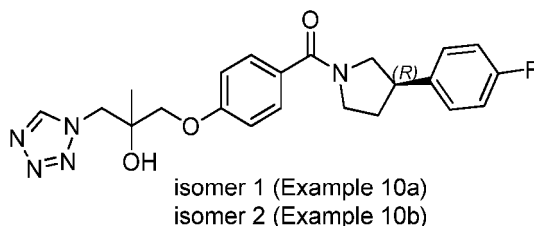


[0307] To a solution of (R)-4-(2-hydroxy-3-(2H-tetrazol-2-yl)propoxy)benzoic acid (intermediate **57**, 50 g, 0.189 mol), (R)-3-(4-fluorophenyl)pyrrolidine hydrochloride (intermediate **3b**, 45.8 g, 0.227 mol) and DIPEA (SIGMA-ALDRICH, 99 mL, 0.568 mol) in anh. DMF (1.9 L) at 0°C under nitrogen atmosphere was added COMU (ALFA AESAR, 105 g, 0.246 mol) and the reaction mixture was stirred at 0°C for 1h and then at rt overnight. The mixture was quenched by the dropwise addition of water (1 L), 1 M HCl (0.5 L) and sat. NaCl (0.5 L). The resulting mixture was then partitioned with EtOAc (2 L), the phases were separated and the aqueous one was re-extracted with EtOAc (1 L). The organic layers were combined and washed sat. aq. NH₄Cl (1 L), sat. aq. Na₂CO₃ (1 L) and brine. The organic layer was dried over anh. Na₂SO₄, filtered and evaporated to dryness under vacuum to obtain 125 g of crude material. The crude was purified by chromatography on silica gel column and eluted with CyHex:EtOAc/EtOH (3/1) (from 100:0 to 50:50). The desired fractions were collected and the solvent was removed in vacuo to obtain the title compound impurified with by-products derived of COMU. For remove these impurities, the product was dissolved in EtOAc (1 L) and washed with 2 M HCl (×3) and sat. aq. Na₂CO₃ sol. (×1) and brine. The organic layer was dried over anh. Na₂SO₄, filtered and evaporated to dryness under vacuum to obtain 75 g of ((R)-3-(4-fluorophenyl)pyrrolidin-1-yl)(4-((R)-2-hydroxy-3-(2H-tetrazol-2-yl)propoxy)phenyl) methanone.

[0308] ((R)-3-(4-fluorophenyl)pyrrolidin-1-yl)(4-((R)-2-hydroxy-3-(2H-tetrazol-2-yl)propoxy)phenyl) methanone (75 g) was dissolved in acetone (0.250 L) and Et₂O (1 L) was slowly added. After 24 h of crystallization, the solid was filtered to give ((R)-3-(4-fluorophenyl)pyrrolidin-1-yl)(4-((R)-2-hydroxy-3-(2H-tetrazol-2-yl)propoxy)phenyl)methanone (49 g, 0.119 mol, 62.9% yield, with an ee > 97%). The mother liquor was concentrated to obtain ((R)-3-(4-fluorophenyl)pyrrolidin-1-yl)(4-((R)-2-hydroxy-3-(2H-tetrazol-2-yl)propoxy)phenyl)methanone (26 g, 63.2 mmol, 33.4% yield, with a ee ≈ 95%). ¹H NMR (300 MHz, DMSO-d₆) δ ppm: 8.97 (s, 1H), 7.55 (d, J= 8.6 Hz, 2H), 7.43-7.27 (m, 2H), 7.21-7.07 (m, 2H), 7.05-6.92 (m, 2H), 5.60 (d, J= 5.3 Hz, 1H), 4.95-4.74 (m, 2H), 4.40 (br.s., 1H), 4.15-4.02 (m, 2H), 3.98-3.72 (m, 1H), 3.71-3.34 (m, 4H), 2.31-2.16 (m, 1H), 2.07-1.87 (m, 1H). [ES+ MS] m/z 412 (MH⁺). Chiral SFC determination using ChiralPak IA 150x3 mm column (CO₂/MeOH (1%DEA) 60:40 (v/v), F=3.0 mL/min, R_t=3.54 min).

Example 8: ((R)-3-(4-Fluorophenyl)pyrrolidin-1-yl)(4-((S)-2-hydroxy-3-(2H-tetrazol-2-yl)propoxy)phenyl)methanone

[0309]



[0315] Examples 10a-b were also prepared using analogous conditions described in Examples 9a-b using 4-(2-hydroxy-2-methyl-3-(1H-tetrazol-1-yl)propoxy)benzoic acid (intermediate **63**, 250 mg, 0.898 mmol). The diastereomer mixture was separated by chiral SFC to afford isomer 1 (123 mg, ee 100%, Example 10a) and isomer 2 (125 mg, ee 100%, Example 10b).

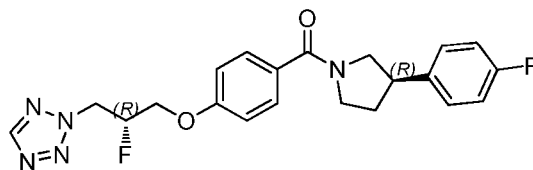
isomer 1 (Example 10a) - First eluted in SFC. ^1H NMR (300 MHz, CDCl_3) δ ppm: 8.84 (d, J = 5.5 Hz, 1H), 7.58-7.48 (m, 2H), 7.30-7.15 (m, 2H), 7.12-6.99 (m, 2H), 6.90-6.80 (m, 2H), 4.76-4.57 (m, 2H), 4.16-3.32 (m, 8H), 2.47-2.27 (m, 1H), 2.21-1.94 (m, 1H), 1.35 (d, J = 6.0 Hz, 3H). [ES+ MS] m/z 426 (MH^+).

isomer 2 (Example 10b) - Second eluted in SFC. ^1H NMR (300 MHz, CDCl_3) δ ppm: 8.84 (d, J = 5.5 Hz, 1H), 7.58-7.48 (m, 2H), 7.30-7.15 (m, 2H), 7.12-6.99 (m, 2H), 6.90-6.80 (m, 2H), 4.76-4.57 (m, 2H), 4.16-3.32 (m, 8H), 2.47-2.27 (m, 1H), 2.21-1.94 (m, 1H), 1.35 (d, J = 6.0 Hz, 3H). [ES+ MS] m/z 426 (MH^+).

[0316] Chiral SFC determination using ChiralPak AD-H 150x3 mm column (CO_2/MeOH 60:40 (v/v), F = 2.0 mL/min, R_t isomer 1 = 3.26 min, R_t isomer 2 = 5.34 min).

Example 11: (4-((R)-2-Fluoro-3-(2H-tetrazol-2-yl)propoxy)phenyl)((R)-3-(4-fluorophenyl)pyrrolidin-1-yl)methanone)

[0317]



[0318] Diethylaminosulfur trifluoride (DAST, SIGMA-ALDRICH, 65 μL , 0.492 mmol) was added to a solution of ((R)-3-(4-fluorophenyl)pyrrolidin-1-yl)(4-((S)-2-hydroxy-3-(2H-tetrazol-2-yl)propoxy)phenyl)methanone (Example **8**, 100 mg, 0.243 mmol) in DCM (3 mL) at 0°C . The mixture was stirred at rt for 5 h. Then, a new addition of diethylaminosulfur trifluoride (SIGMA-ALDRICH, 32.1 μL , 0.243 mmol) was done and the mixture was stirred at rt overnight. The reaction was quenched with sat. NaHCO_3 solution (5 mL), diluted with DCM (5 mL) and extracted three times with DCM (10 mL). The combined organics were dried over Na_2SO_4 , filtered and concentrated under reduced pressure. The crude was purified by chromatography on silica gel and eluted with CyHex:EtOAc/EtOH (3/1) (from 100:0 to 65:35) to obtain the title compound (42 mg, 41.8% yield, ee 98.5%) as a white solid. ^1H NMR (300 MHz, $\text{DMSO}-d_6$) δ ppm: 9.06 (s, 1H), 7.57 (d, J = 8.6 Hz, 2H), 7.43-7.28 (m, 2H), 7.21-6.98 (m, 4H), 5.54-5.31 (m, 1H), 5.30-5.14 (m, 2H), 4.54-4.23 (m, 2H), 3.99-3.34 (m, 5H), 2.37-2.15 (m, 1H), 2.10-1.87 (m, 1H). [ES+ MS] m/z 414 (MH^+). Chiral SFC determination using Chiral Pak IC 150x3 mm column (CO_2/MeOH (1%DEA) 70:30 (v/v), F = 2.0 mL/min, R_t = 5.06 min).

Example 12: (4-((S)-2-Fluoro-3-(2H-tetrazol-2-yl)propoxy)phenyl)((R)-3-(4-fluorophenyl)pyrrolidin-1-yl)methanone)

[0319]

